



Assignment of bachelor's thesis

Title: Reinforcement Learning for Prší Card Game
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Study program: Informatics
Branch / specialization: Artificial Intelligence 2021
Department: Department of Applied Mathematics
Validity: until the end of summer semester 2026/2027

Instructions

Reinforcement learning in combination with deep neural networks is achieving significant performance in various tasks. The aim of this thesis is to focus on playing the "Prší" card game (similar to its international counterpart, Uno) using deep reinforcement learning methods.

Detailed description:

- Design and implement an appropriate framework that enables self-play as well as interaction of an agent with real players.
- Research the recent reinforcement learning methods and select some basic and at least two deep learning-based methods that should be suitable for the given task.
- Implement all the selected methods that will primarily learn by playing with itself (self-play).
- Assess the quality of the learned models and test them against real players. Discuss the results.